

NONCONSTRUCTIVE ADVANCES IN POLYNOMIAL-TIME COMPLEXITY

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The field of computational complexity for concrete, practical combinatorial problems has developed in a remarkably smooth fashion. One can point to several features of the theory of polynomial-time computability which make it especially well-behaved, including: (1) the modelling of feasible computing by polynomial-time complexity is well-supported by the fact that almost all known polynomial-time algorithms for natural problems have running times bounded by polynomials of small degree; (2) problems are invariably known to be decidable in polynomial time by direct evidence in the form of efficient algorithms; (3) while the theory is formulated in terms of decision problems, almost all known algorithms proceed by actually constructing a solution to the problem at hand.

Herein we illustrate how recent advances in graph theory and graph algorithms dramatically alter this situation on all three counts. Powerful and easy-to-apply tools are now available for classifying problems as decidable in polynomial time by nonconstructively proving only the existence of polynomial-time decision algorithms. These tools neither specify the degree of the polynomial, nor produce the decision algorithm, nor guarantee that such an algorithm is of any use in constructing a solution. These developments present both practitioners and theorists with novel challenges.

Keywords: Analysis of algorithms, combinatorial problem, computational complexity, theory of computation

1. Introduction

Until recently, it seemed reasonable to suggest that “the polynomially solvable problems that arise naturally tend to be solvable within polynomial time bounds that have degree 2 or 3 at worst and that do not involve extremely large coefficients” [2, p. 9]. If this statement is indeed correct, then algorithm designers now face a number of new, interesting challenges. Advances [10,11] in proving a long-standing conjecture by Wagner [14] and in solving the disjoint paths problem have suddenly made available powerful methods which can identify many natural problems as solvable in

polynomial time by insuring only the *existence* of algorithms with polynomial-time bounds.

While the graph-theoretical techniques employed in these new advances are deep and intricate, the results of immediate importance to concrete complexity theory are easy to state and apply. We shall illustrate their utility in proving membership in P, even for problems that are *not* directly about graphs.

2. Background

A graph H is a *minor* of a graph G , written $H \leq G$, if a graph isomorphic to H can be obtained from a subgraph of G by contracting edges. For example, the construction depicted in Fig. 1

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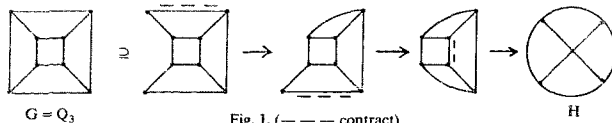


Fig. 1. (--- contract).

shows that the graph of the wheel with four spokes is a minor of the graph of the three-dimensional binary cube.

Note that the relation $\leq$ defines a partial ordering on finite graphs. A family F of finite graphs is said to be *closed* under the minor ordering if the two conditions G in F and $G \geq H$ together imply that H must be in F .

Theorem A ([10]). *Any family of finite graphs which is closed under the minor ordering and which excludes some planar graph has a polynomial-time membership test.*

The proof of Theorem A is based on guaranteeing that the complement of such a family contains a finite number of minimal elements in the minor ordering. We shall henceforth refer to this minor-minimal collection of graphs as the *obstruction set* for family F . Therefore, we have the following characterization of F : G is in F if and only if there exists no H in F 's obstruction set such that $H \leq G$. The polynomial-time bound for the membership test follows from the additional guarantee that whether $H \leq G$ can be determined in polynomial time.

Interestingly, the proof of Theorem A depends on a form of the Axiom of Choice: (It is conceivable, but presently not known, that it may in fact be equivalent to a form of the Axiom of Choice, as many mathematical statements are [12].) Hence, when we can apply Theorem A, we are assured of a finite obstruction set without being given any information about how to find the set, how large it is, or even the size of the largest graph it contains. That is, this information is available neither from the proof of Theorem A nor from the manner in which we employ it. Thus, we have available, for the first time, a general and powerful (but highly nonconstructive) tool to show polynomial-time decidability.

3. A simple graph problem

We shall now illustrate the ease with which Theorem A can be applied to prove polynomial-time decidability. We know of no alternate proof that this problem is in P .

A graph G is said to be *outerplanar* if it can be embedded in the plane minus an open disk in such a way that every vertex of G lies on the boundary of the disk. The *disk dimension* of a planar graph G , important in such applications as printed circuit board routing, is defined to be the least integer k such that G embeds in the plane minus k open disks, with every vertex of G on a boundary of one of the removed disks. Formally,

DISK DIMENSION (DD)

INSTANCE: A planar graph G and an integer k .

QUESTION: Is the disk dimension of G less than or equal to k ?

3.1. Theorem. *For every fixed k , DD is solvable in polynomial time.*

Proof. The family of graphs for which $DD(G, k) = \text{"yes"}$ is closed under the minor ordering since a sequence of edge and vertex deletions followed by a sequence of edge contractions can never increase the number of disks required. For each k , it is trivial to construct a planar obstruction (that is, a planar graph G for which $DD(G, k) = \text{"no"}$). We can therefore apply Theorem A. $\square$

4. An important nongraph problem

A promising, semi-custom VLSI layout style called *gate matrix* was introduced in [9] for CMOS circuits. The combinatorial problem of interest takes as input a collection of nets (rows) $N = \{N_1, \dots, N_n\}$ and their respective connections to a

set of gates (columns) $G = \{G_1, \dots, G_m\}$. We seek to know whether there exists a permutation of the gates which allows the circuit to be laid out in k or fewer tracks. Formally,

GATE MATRIX LAYOUT (GML)

INSTANCE: A Boolean matrix M and an integer k .

QUESTION: Can we permute the columns of M so that, if in each row we change to * every 0 lying between the row's leftmost and rightmost 1, then no column contains more than k 1s and *s?

A 0 that is changed to a * is termed a 'fill-in', and represents the fact that all gates in a net must be physically connected. If, for a given gate permutation, a pair of nets do not overlap, then they can share a track. Thus, minimizing the maximum number of 1s and *s in any column, over all permutations, corresponds to minimizing the number of tracks and hence the area needed for a GML chip.

Interestingly, GML as defined above turns out (with an appropriate transposition of rows and columns) to be identical to the combinatorial problem faced in minimizing the chip area in other important VLSI layout styles as well, including that of *multiple PLA folding*, *Weinberger arrays*, and *one-dimensional logic arrays* [5]. Hence, as a bonus, the results which follow apply to these styles as well.

It is known from [6] that GML is NP-complete. Accordingly, the significance of the problem is reflected in the continuing investigation of fast heuristic algorithms (see, for example, [1,4,7,8,15]). Despite the intense interest in GML, and the equivalent layout formulations mentioned above, no published heuristic has been certified as even a relative approximation algorithm [3], and it is known that no absolute approximation algorithm is possible unless $P = NP$ [1].

Surprisingly, however, we shall now proceed to prove that GML is in fact in P for any fixed k . Our first task is to devise a way to transform an instance of GML into a suitable representation as a graph. Given a Boolean matrix M , we map M to an 'expanded' form $x(M)$ in which any column i with $j > 2$ 1s is replaced with $\binom{j}{2}$ columns, each

with two 1s, representing every distinct way to choose a pair of 1s from column i . For example, if

$$M = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix},$$

then

$$x(M) = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}.$$

Notice that this notion of matrix expansion is many-to-one and hence is not invertible.

4.1. Lemma. $GML(M, k) = GML(x(M), k)$.

Proof. Suppose $GML(M, k) = \text{"yes"}$. Then there is some permutation p of the columns of M so that, after *s are included, no column of $p(M)$ contains more than k 1s and *s. Consider the matrix $x(p(M))$, the expanded form of $p(M)$. For a column i of $p(M)$ with $j > 2$ 1s, each of the $\binom{j}{2}$ columns replacing column i in $x(p(M))$ can incur a * only where column i had either a 1 or a *. Furthermore, columns to the left and right of column i are unaffected by its expansion. Therefore, $GML(x(p(M)), k) = \text{"yes"}$. Since $x(p(M)) = p'(x(M))$ for some appropriate permutation p' of the columns of $x(M)$, we have $GML(x(M), k) = \text{"yes"}$.

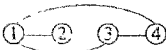
Suppose $GML(x(M), k) = \text{"yes"}$. Then there is some permutation p of the columns of $x(M)$ so that, after *s are included, no column of $p(x(M))$ contains more than k 1s and *s. Consider any column i of M with $j > 2$ 1s. Let f and ℓ denote the indices of the first and last columns, respectively, from column i 's $\binom{j}{2}$ expanded columns in $p(x(M))$. Since all nets from column i are pairwise connected, no two can share a track, implying that one of the $\binom{j}{2}$ expanded columns has 1s and *s in every net of column i . Let h denote the index of such a column in $p(x(M))$, where $f \leq h \leq \ell$. Now replace column h of $p(x(M))$ with column i of M . The new column has more 1s than did the old, but incurs fewer *s so that its total after *s are included is unchanged. Next, delete from $p(x(M))$

the $(\frac{1}{2}) - 1$ other columns expanded from column i of M . The nets of column i have not been stretched across any additional columns by this procedure, and thus any remaining columns between where columns f and l' originally were incur no new $*s$. We conclude that the resulting matrix requires no more than k tracks. Iterating this construction yields a permutation p' of the columns of M , and hence $GML(M, k) = \text{"yes"}$. $\square$

Therefore, from now on, we need only consider $x(M)$. Since each column of $x(M)$ contains at most two 1s, we map $x(M)$ to a graph $g(x(M))$ whose vertices correspond to nets and whose edges represent columns. For example, if

$$x(M) = \begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix},$$

then $g(x(M))$ is the graph



Notice that g is invertible in that we can recover from $g(x(M))$ the set of columns in $x(M)$. (Also, $g(x(M))$ may contain self-loops and multiple edges.) Of importance in what follows is that we need only consider graphs, since for any graph G there exists at least one Boolean matrix M such that $G = g(x(M))$.

4.2. Lemma. *The family of graphs for which $GML(G, k) = \text{"yes"}$ is closed under the minor ordering.*

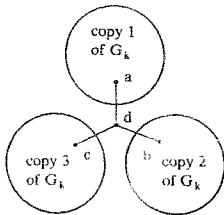


Fig. 2.

Proof. Suppose M_1 and M_2 are Boolean matrices such that $G_1 = g(x(M_1))$ is a minor of $G_2 = g(x(M_2))$. Furthermore, suppose $GML(G_2, k) = \text{"yes"}$. Discarding edges (and, possibly, vertices) from G_2 to form a subgraph corresponds to eliminating columns (and, possibly, rows) from $x(M_2)$. Contracting an edge corresponds to deleting a column and replacing a pair of rows with a single row containing their element-by-element Boolean OR. Neither operation can increase the number of tracks required in the layout. Therefore, $GML(G_1, k) = \text{"yes"}$. $\square$

4.3. Lemma. *For any fixed k , there exists a planar graph G such that $GML(G, k) = \text{"no"}$.*

Proof. We proceed by induction on k . It is trivial to verify that $GML(K_3, 2) = \text{"no"}$. Suppose there exists a planar, connected graph G_k such that $GML(G_k, k) = \text{"no"}$ for some $k \geq 2$. Then we use the construction depicted in Fig. 2 to define a (planar) graph G_{k+1} where the vertices a , b , and c are selected arbitrarily from their respective copy of G_k . Let $p(G_{k+1})$ denote a minimum-track permutation of the (expanded) matrix corresponding to G_{k+1} . Clearly, $p(G_{k+1})$ contains a column from copy 1 of G_k with at least $(k+1)$ 1s and $*s$. Let i_1 denote the index of this column. Similarly, we define columns i_2 and i_3 from copies 2 and 3 of G_k , respectively. Without loss of generality, we assume $i_1 < i_2 < i_3$. If the three columns denoting the edges $\{(a, d), (b, d), (c, d)\}$ do not all either precede or follow column i_2 , then vertex d forces column i_2 to incur an additional $*$. In the former case, the fact that vertex c is connected to the vertices represented in column i_3 forces an additional $*$ in column i_2 . In the latter case, the fact that vertex a is connected to the vertices represented in i_1 does likewise. Therefore, in any event, column i_2 requires at least $(k+2)$ 1s and $*s$, and $GML(G_{k+1}, k+1) = \text{"no"}$. $\square$

4.4. Theorem. *For every fixed k , GML is solvable in polynomial time.*

Proof. Lemma 4.1 and the invertability of g insure that, for a Boolean matrix M , it is sufficient to consider the graph $g(x(M))$ whose construction takes at most $O(m^2)$ time. From Lemmas 4.2 and

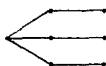


Fig. 3.

4.3 we are able to apply Theorem A and guarantee that, for any fixed k , there is a decision algorithm whose time complexity is bounded by a polynomial in the size of m and n . $\square$

We note that the fixed- k version of GML is still a very difficult problem, not to be confused with certain combinatorial problems that are 'easy' in the sense that their fixed-parameter versions can be solved by brute-force methods that merely check a predetermined list of candidate solutions. The CLIQUE problem [2] is, for example, such a problem since we can check for a clique of size k by considering each of the $\binom{n}{k}$ subgraphs of size k for any graph on n vertices in $O(n^k)$ time. Instances of GML exist, however, with but two satisfactory permutations and $m! - 2$ unsatisfactory ones [1], preventing any efficient brute-force attack which focusses on a predetermined set of column permutations.

Incidentally, encouraged by the guarantee provided by Theorem 4.4, we have begun to close in on the obstructions and hence the polynomial-time algorithms for fixed- k GML, at least for small k . For example, we know that for $k=2$ the only obstructions are K_3 and the graph depicted in Fig. 3.

Furthermore, although it is beyond the scope of this brief exposition, we can also show that GML is *self-reducible* in that we can efficiently turn decision into construction.

5. Even better tools

Newly announced [13] but as yet unpublished progress on proving Wagner's conjecture and on solving the disjoint paths problem is expected to provide even more powerful methods for determining polynomial-time decidability. In particular: (1) the requirement for a planar obstruction

is eliminated, (2) a family of graphs need only be minor-closed within a polynomial-time recognizable set of graphs, and (3) the resulting decision algorithm runs in $O(v^3e)$ time where v and e denote the respective number of vertices and edges in the problem graph. Of course, this method remains highly nonconstructive. There is still neither a general procedure for finding the decision algorithm nor a guarantee that a "yes" decision can be transformed into an acceptable solution to the problem at hand.

This new situation immediately begs many more intriguing questions. For example, these reported results make it easy to show that KNOTLESSNESS (given a graph G , is there an embedding of G in 3-space such that all cycles of G are unknotted) is in P [16], even though the problem is not otherwise even known to be decidable!

We conclude that the class of problems which are solvable in deterministic polynomial time appears to be much richer and more interesting than previously thought. Nonconstructive membership tools now available raise a number of important issues in the study of algorithm design whose resolution, if that is possible, should be extremely enlightening.

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Reference added in proof